

DHAIRYAJEET SINGH

AI/ML Engineer | Full-Stack AI Developer

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PROFESSIONAL SUMMARY

Computer Science undergraduate shipping production AI systems across computer vision, NLP, and MLOps. Deployed a real-time AI proctoring platform, a marketing intelligence engine analyzing 1,000+ data points, and automation tools with measurable ROI. Skilled across the full stack: model development, backend engineering (FastAPI, Flask), and cloud deployment (Azure, Docker, Nginx, CI/CD).

EXPERIENCE

AI/ML Engineer Intern - Knovon Consulting, Delhi, India

June 2025 – Dec 2025

- Shipped production **real-time AI proctoring system** for [TalkReady](#) interview platform, deployed via FastAPI on Azure VM with HTTPS (Certbot); system now used by recruiters to securely evaluate candidates at scale.
- Engineered a multi-model **computer vision + voice biometric pipeline**: Face Verification (**93%**, Buffalo_L), Mobile/Face Detection (87%, YOLOv11m), Anti-Spoofing (83%), and Voice Verification (Encapa) — hardening candidate identity against spoofing.
- Built a production **resume parsing pipeline** (OCR + text extraction) with PyTesseract, PDFPlumber, and PyPDF2, supporting multiple formats; live in production.
- Developed a full-stack **bulk email automation tool** with Microsoft OAuth, CSV variable mapping, and org-wide mail support — replacing manual outreach workflows.
- Owned end-to-end **DevOps**: Dockerization, CI/CD (GitHub Actions), Nginx, Azure VM provisioning, VPN setup, and cloud infrastructure monitoring across internal and client projects.

Smart Manufacturing Intern - UNO Minda, Manesar, India

Jul 2024 – Aug 2024

- Built a Python + Pandas **data automation tool** that cut manual data entry by **50%**, saving the manufacturing team **60+ hours per month**.
- Developed an **LSTM time-series model** predicting machine running costs with **80% accuracy**, enabling proactive maintenance planning on the factory floor.

Data Analyst Intern - To-Let-Globe, Delhi, India (Remote)

May 2024 – Aug 2024

- Analyzed rental market data across **20+ Lucknow neighborhoods** — data cleaning, EDA, and visualization — surfacing pricing trends that improved client decision-making efficiency by **40%**.

PROJECTS

Nimbus AI - Marketing Intelligence Platform

Aug 2025

Python, BART, LLaMA 3, Web Crawlers, Fine-Tuning, Vector Embeddings

- Engineered an AI-driven platform analyzing **1,000+ competitor strategies**, reducing client competitive research time by **75%**.
- Built a recommendation engine that lifted client marketing ROI by **37%** on average by surfacing 15+ strategy gaps versus competitors.
- Fine-tuned BART and LLaMA 3 models on crawler-sourced data, achieving **89% relevance** across 500+ competitor data points.

Text-to-3D Generation Web App

June 2025

Shape-E, Flask, Python, PLY, GitHub PAT, Mesh Processing

- Built a web app converting natural-language prompts into downloadable **.ply 3D models** using Shape-E with a Flask backend, secure GitHub PAT handling, and an optimized mesh-processing + visualization pipeline.

AI News Letter Agent

- Developed an autonomous multi-agent workflow for newsletter generation, integrating planning, web research, content writing, critique, and publishing.
- Designed a self-improving feedback loop with HITL approval and automated email distribution, improving content reliability and factual accuracy.

Facial Emotion Recognition (FER)

Oct 2024

TensorFlow, Keras, CNN, LSTM, FER-2013 Dataset, Google Colab

- Trained a hybrid **CNN + LSTM model** on the FER-2013 dataset for emotion classification, achieving **75.76% training accuracy** and 63.22% validation accuracy through data augmentation and hyperparameter optimization.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SQL, Bash

AI / ML: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, LLMs, GANs, Fine-Tuning, RAG, PyTorch, TensorFlow, Keras, Scikit-Learn, Hugging Face Transformers, LangChain, LangGraph, Ultralytics (YOLO), InsightFace, ONNX, Ollama, spaCy, NLTK

Backend & Data: FastAPI, Flask, REST APIs, async programming, Streamlit, Web Sockets, JWT Authentication, PostgreSQL, DBMS, Networking, Pandas, NumPy, Matplotlib, Seaborn, BeautifulSoup, MySQL, SQLAlchemy,

DevOps & Tools: Azure (VMs, Resource Groups), Docker, Kubernetes, Nginx, GitHub Actions, CI/CD, Linux, VPN, Certbot/HTTPS, Git, Postman

EDUCATION & ACHIEVEMENTS

B.Tech, Computer Science - GTBIT, GGSIPU, Delhi CGPA: 8.6/10

Sept 2023 – July 2027

Achievements: [3rd Place, Smart Delhi Ideathon \(SDI\) 2025](#) • [3rd Place, IC Hack 2.0](#) inter-college coding hackathon by IEEE Delhi Section.